Examiner only

2. A balloon of volume $1.60 \times 10^{-3}\,\text{m}^3$ contains helium at a pressure of $1.20 \times 10^5\,\text{Pa}$ and a temperature of 293 K. (Relative molecular mass of helium = 4.)

(a) Determine:

(i) the number of molecules in the balloon; [2]

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(ii) the rms speed of the molecules. [2]

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(b) The product of the pressure and the volume of the gas is given by $pV = nRT$ and also by $pV = \frac{1}{3}Nm\overline{c^2}$.

(i) Show that the mean kinetic energy of a gas molecule is $\frac{3}{2}kT$ where k is the Boltzmann constant. [4]

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Examiner only

(ii) Calculate the mean kinetic energy of a molecule in the gas. [1]

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(iii) Determine the internal energy of the gas in the balloon. [1]

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(c) (i) Explain, in terms of the molecules, the difference in the nature of the **internal energy** in an ideal gas and in a liquid. [3]

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(ii) State the temperature of a system for its internal energy to be a minimum. [1]

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